Guide: Where to Look for an Unbounded Orbit

Collatz proof project

September 5, 2026

What Lean now proves

Every hypothetical unbounded positive Collatz orbit has a point from which its multiplicative coefficient never drops below one. Such starting points occur arbitrarily late. Conversely, a positive seed whose coefficient never drops below one must have an unbounded orbit.

This turns the escape problem into one exact task: prove that every positive seed eventually has a coefficient below one. That task is still open. Even after solving it, nontrivial cycles would still need to be excluded.

Why a suitable starting point exists

The earlier summability result proves that the inverse coefficient $2^t/3^{j_t}$ tends to zero along any unbounded orbit. Its value at time zero is one, so its largest value is attained at some finite time.

Start at that maximum. Every later inverse coefficient, measured relative to the new starting point, is at most one. This is precisely the required noncontraction condition. The same argument can start after any prescribed time; it does not say that every late point has the property.

For the converse, suppose such a seed had a bounded orbit. The exact orbit identity would then bound its ideal correction too. But bounded ideal correction already forces escape, giving a contradiction.

A first arithmetic restriction

A seed with no coefficient contraction must be odd. If it is also two modulo three, then $(2N-1)/3$ is a smaller positive odd predecessor. Its first coefficient is $3/2$, so it also has no future contraction.

Therefore a smallest positive seed with this property cannot be two modulo three. This does not eliminate the remaining residues. Lean also proves a general rule for transporting noncontraction backward through a finite prefix whose coefficients all stay at least one.

What this changes, and what it does not

The reduction gives complete coverage of hypothetical unbounded positive orbits by a specific arithmetic class. We still have to prove that this class has no positive natural seed. Passing a finite prefix test cannot establish the infinite condition.

The earlier 65-step descent certificate helps when a coefficient contraction happens within that horizon. It does not prove that every seed contracts within 65 steps, or at any time. The Collatz goal remains open; no mathematical priority claim is made here.

Verification

The build and 139-declaration standard-foundations audit pass. Run `lake build Collatz.NoncontractingTail` from `lean/`.

The technical paper gives the exact equivalences and hypotheses.